

DESIGN THE
INTERNET

C2

T3

REQUIREMENTS

01

AVAILABILITY

Ensures the network is reachable by any user, anywhere, at any time through a decentralized "network of networks"

02

RELIABILITY

Prevents data loss during crashes using fail-safe mechanisms like Backup Parents and retransmissions

03

INTEROPERABILITY

Allows different hardware and operating systems to communicate using a common, standardized data packet format

04

SCALABILITY

Maintains performance during growth by using a hierarchical addressing scheme that simplifies global routing

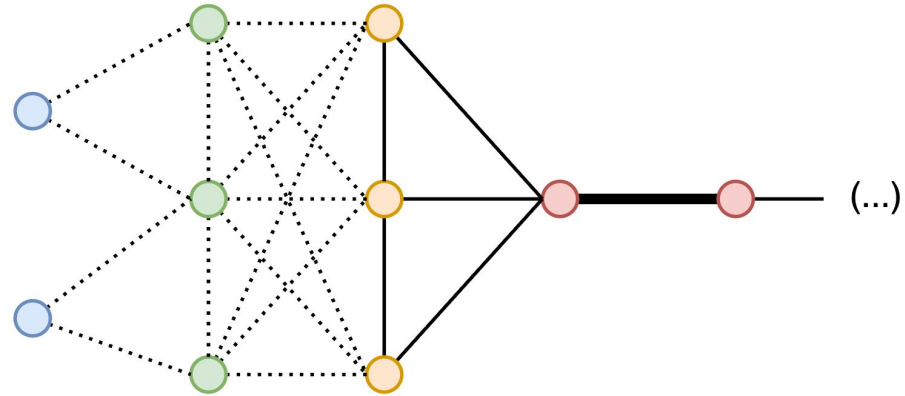
05

SECURITY

Protects privacy and data integrity through end-to-end encryption and digital signatures to ensure user trust

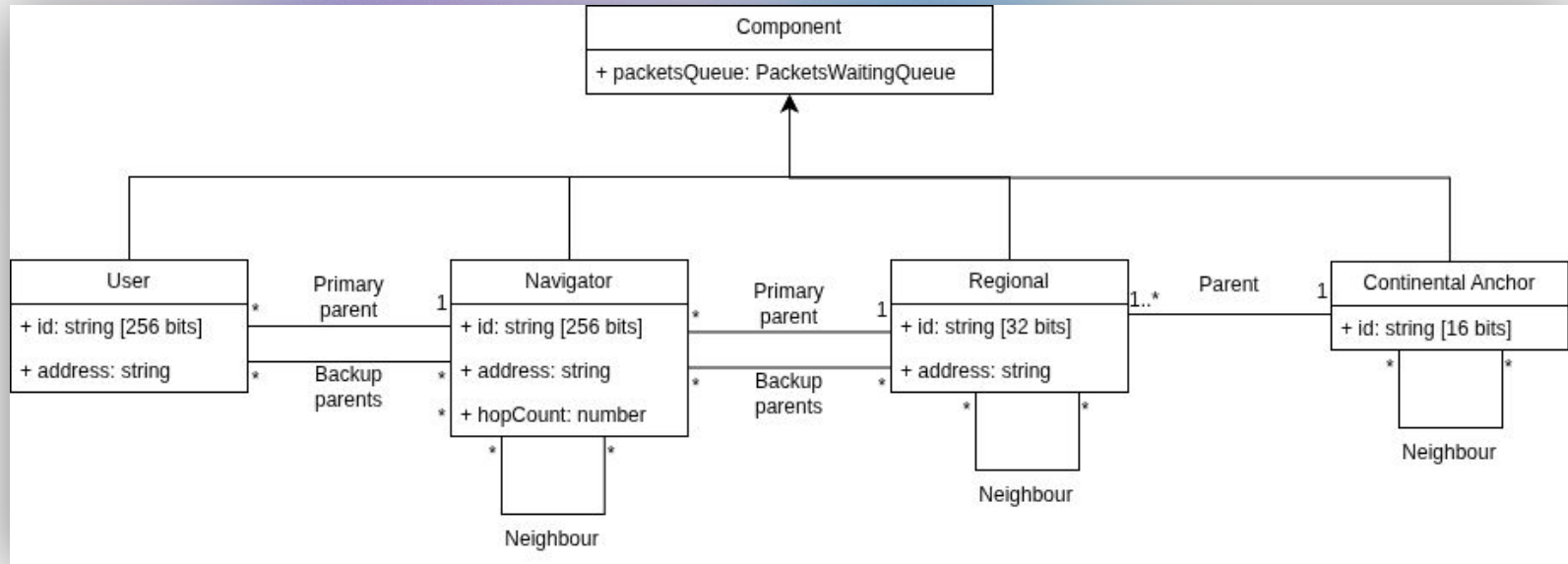
ARCHITECTURE OF THE INTERNET

Our design uses a tiered
'network of networks' to balance
worldwide reach with local stability



- Continental Anchor
- Regional
- Navigator
- User
- Super-Fiber
- Fiber
- Wireless

RELATIONS BETWEEN COMPONENTS



DISCOVERY AND DATA ROUTING

ADDRESSING

Every node is assigned a unique, hierarchical address **[Anchor].[Regional].[Navigator].[User]**

DISCOVERY

New users "shout" to find nearby Navigators, while Navigators "shout" to find neighbors and Regionals.

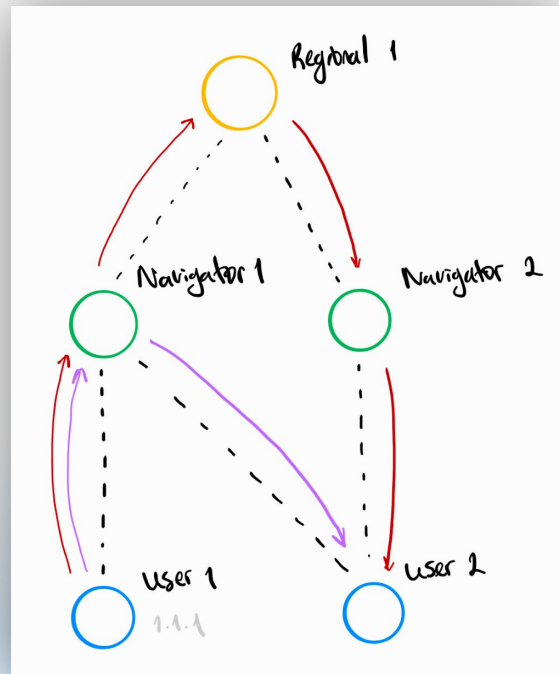
ROUTING LOGIC

CLIMBING

Packets "climb" toward the nearest Regional by always hopping to the neighbor with the lowest internal Hop Count.

SHORTCUTS

If a Navigator recognizes a local ID, it skips the core and delivers the packet directly to the recipient.



RESILIENCE AND TRAFFIC CONTROL

FAILOVER SAFETY NET

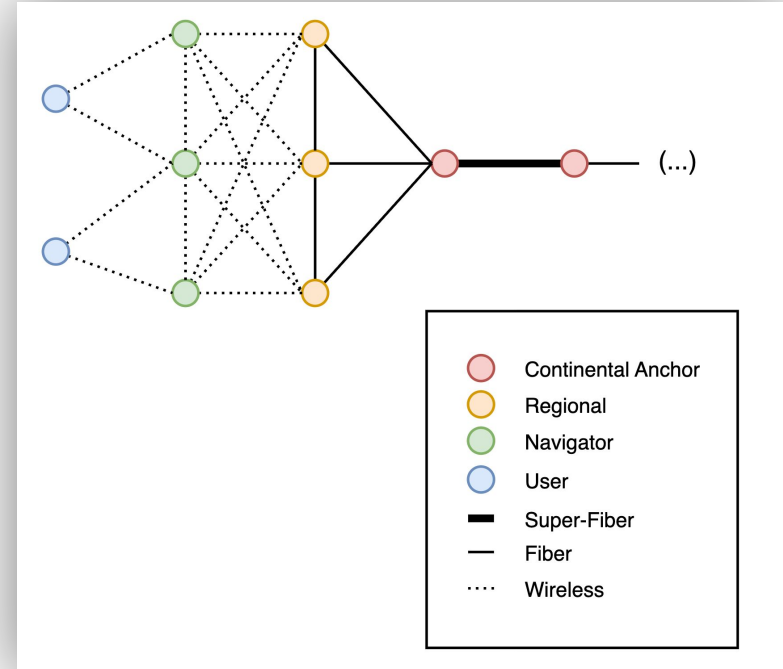
Users and Navigators maintain a “Backup Parents” list. If a primary connection fails, the device instantly "hops" to an alternative route to prevent a break.

RELIABILITY MECHANISM

Data packets include sequence numbers and acknowledgments (ACKs) so end users can detect missing data and request retransmission.

PRIORITY SYSTEM

Nodes use a waiting queue to process “Emergency” and “Real-time” traffic before background data to prevent “traffic jams”.



OUR DESIGN

AMBITIOUS COMPLEXITY: Built-in network intelligence to manage traffic and connections.

PRIORITY SYSTEM: Actively manages data based on importance (e.g., Emergency vs. Normal).

BACKUP PARENTS: Provides instant “hops” to alternative Navigators to prevent connection breaks.

REAL INTERNET

ESSENTIAL SIMPLICITY: A “modest” and “unintelligent” core designed for maximum scale.

BEST-EFFORT MODEL Makes no performance guarantees; all packets are treated equally.

ASSUMES FAILURE: Treats dropped packets as normal and relies on host devices to fix the mess.

LESSON LEARNED

*A “dumb” network is often more “intelligent”
for growing at a global scale.*